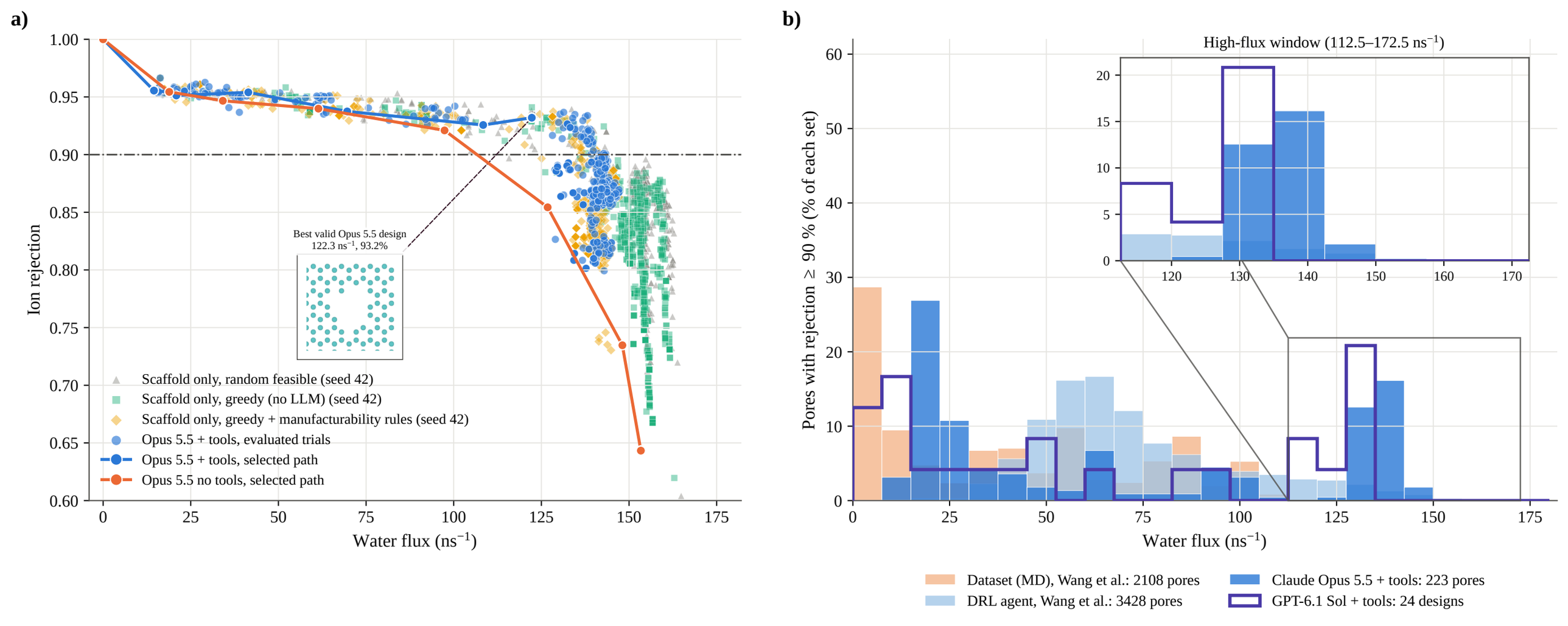

a)
b)
Ion rejection
Water flux (ns^{-1})
Best valid Opus 5.5 design
122.3 ns^{-1}, 93.2%
Scaffold only, random feasible (seed 42)
Scaffold only, greedy (no LLM) (seed 42)
Scaffold only, greedy + manufacturability rules (seed 42)
Opus 5.5 + tools, evaluated trials
Opus 5.5 + tools, selected path
Opus 5.5 no tools, selected path
Pores with rejection ≥ 90 % (% of each set)
High-flux window (112.5–172.5 ns^{-1})
Dataset (MD), Wang et al.: 2108 pores
DRL agent, Wang et al.: 3428 pores
Claude Opus 5.5 + tools: 223 pores
GPT-6.1 Sol + tools: 24 designs